1-to-1 Correspondence: The Concrete Step Counter

Numbers to 5 & 10 • Standard: CCSS.MATH.K.CC.B.4

BOOK STORY OPENER: How Ten Fingers & an Empty Shell Created Place Value & The 1-to-10 Staircase

Long before paper or keyboards existed, ancient shepherds counted flocks with clay pebbles ('calculi' in Latin—the root of 'calculator'!) and Maya astronomers drew a shell for zero so an empty tray had its own symbol. Dropping 1 counter at a time into the Ten-Frame while pressing [1] [+] builds the exact 1-to-10 Staircase from Chapter 3 of From Blocks to the Stack.

Companion Book: From Blocks to the Stack (SC-6 Manual) — Ch. 3: Counting, Subitizing & Place Value

1. CONCRETE MANIPULATIVE SETUP

Manipulative Kit: Ten-Frame Tray & Counting Pegs

Place 1 peg into the first well of the Ten-Frame. Have students observe the single object.

2. STUDENT PREDICTION (ASK BEFORE PRESSING KEYS)

When we place 1 more peg into the next well, how many pegs will be on our frame? (Prediction: 2)

3. CALCULATOR ACTION (SC-6 KEYCAPS & STACK TRACE)

Keystroke Flow: [1] [ENTER] [1] [+] [1] [+]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
[1]	1	First peg placed: 1 enters register X
[ENTER]	1	1 copied to register Y: holds the current count
[1]	1	Second peg placed: 1 enters register X
[+]	2	Press [+]: 1 + 1 = 2! Screen verifies 2 pegs on board
[1]	1	Third peg placed in frame: 1 enters X
[+]	3	Press [+]: 2 + 1 = 3! Frame and screen match perfectly

4. MATHEMATICAL EXPLANATION & RPN ADVANTAGE

Each physical counter corresponds to a single increment. The calculator's stack preserves the running count in Y while new items enter X.

Core Investigation Problem:

Place 4 pegs on your Ten-Frame. Add 1 more peg. Write the keystrokes to show 4 + 1 on the calculator.

More, Less & Zero: Finding Differences

Comparing Sets & Magnitude • Standard: CCSS.MATH.K.CC.C.6

1. FOLLOW-UP EXTENSION SETUP & PREDICTION

Setup: Place 5 pegs in the top row (Row 1) and 2 pegs in the bottom row (Row 2). Prediction: Look at the two rows. Which row has more? How many more pegs are in Row 1 than Row 2?

2. EXTENSION KEYSTROKE SEQUENCE & STACK TRACE

5 ENTER 2 -

Key Pressed	Screen (X)	Internal 4-Level Stack Effect
5	5	Row 1 quantity (5) enters register X
ENTER	5	5 lifted to register Y (first set)
2	2	Row 2 quantity (2) enters register X
-	3	Minus subtracts X from Y: Row 1 has 3 more pegs!

3. TARGET MISCONCEPTIONS & HANDS-ON BLOCK REMEDIATIONS

Misconception 1: Reciting number names faster than placing physical counters (losing 1-to-1 correspondence).

Hands-On Block Remediation: Have the student drop 1 peg into the Ten-Frame well first, say the number aloud, and only then press [1] [+] so the physical peg and Register X stay locked in step.

Misconception 2: Confusing the action of adding 1 more with the total quantity on the tray (cardinality).

Hands-On Block Remediation: Ask the student to sweep their hand around all pegs in the Ten-Frame and compare that total to Register X on the screen.

4. EXTENSION MATHEMATICAL INSIGHT

Comparison subtraction answers 'how many more?'. Register Y holds the larger group and register X holds the smaller group.

Counting, Cardinality & The One-More Step Counter

Name: _____ Date: _____ Class: _____

STUDENT REFERENCE GUIDE: HOW COUNTING MAPS TO THE CALCULATOR

- Every peg placed on the frame matches one count on the calculator screen.
- An empty ten-frame represents zero [0]. Press [C] to start a fresh count.
- To step count by 1: press [1] [ENTER] [1] [+], then press [1] [+] to see the count advance!

Key Sequence: `1` `ENTER` `1` `+` `1` `+`

Problem 1: Place 3 pegs on your frame. Add 2 more. Write the keystrokes to find the total.

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction: _____

Final Answer: _____

[ ] Checked on SC-6

Problem 2: A frame has 5 pegs. If you remove 2 pegs, how many remain?

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction: _____

Final Answer: _____

[ ] Checked on SC-6

Problem 3: Clear the calculator by pressing [C]. What number represents an empty ten-frame?

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction: _____

Final Answer: _____

[ ] Checked on SC-6

Problem 4: Step count by ones starting at 1: [1] [ENTER] [1] [+]. Press [1] [+] two more times. What is X?

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction: _____

Final Answer: _____

[ ] Checked on SC-6

Unit 01: Counting, Cardinality & The One-More Step Counter — Answer Key

Complete solutions and SC-6 verification keystrokes for Pre-K – Kindergarten

1. CORE & EXTENSION INVESTIGATION SOLUTIONS

Core Investigation: Place 4 pegs on your Ten-Frame. Add 1 more peg. Write the keystrokes to show 4 + 1 on the calculator.

Core Solution: 5 pegs. Keystrokes: [4] [ENTER] [1] [+]

Core Check Keys: 4 ENTER 1 +

Extension Challenge: Row 1 has 6 pegs and Row 2 has 2 pegs. How many more pegs in Row 1? Write the keystrokes.

Extension Solution: 4 more pegs. Keystrokes: [6] [ENTER] [2] [-]

Ext. Check Keys: 6 ENTER 2 -

2. STUDENT LAB WORKSHEET SOLUTIONS (PROBLEMS 1–4)

Problem 1: Place 3 pegs on your frame. Add 2 more. Write the keystrokes to find the total.

Expected Answer: 5

SC-6 Verification Keys: 3 ENTER 2 +

Problem 2: A frame has 5 pegs. If you remove 2 pegs, how many remain?

Expected Answer: 3

SC-6 Verification Keys: 5 ENTER 2 -

Problem 3: Clear the calculator by pressing [C]. What number represents an empty ten-frame?

Expected Answer: 0 (Zero)

SC-6 Verification Keys: C

Problem 4: Step count by ones starting at 1: [1] [ENTER] [1] [+]. Press [1] [+] two more times. What is X?

Expected Answer: 4

SC-6 Verification Keys: 1 + 1 +

Hands-On Block & Stack Verification Protocol

Students first build the physical block model and write their mathematical strategy and prediction before verifying their result on the StackCalc calculator.